

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: INTERNET PROGRAMMING
COURSE CODE: CSA4305

COURSE OUTCOME COVERED

CO 4: Formulate data-driven web solutions using XML, XSLT, and JSP within the MVC framework. (L4)

Assessment Tool: Class Test

Weightage: 10%

QUESTIONS:

Total Marks: 20

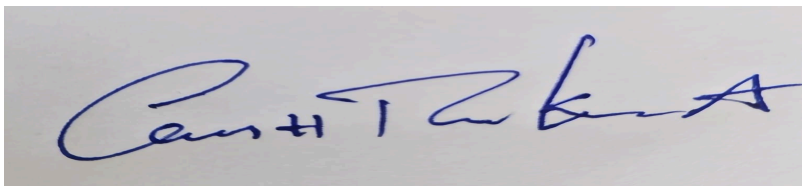
1. Explain the fundamental concepts used in web development by defining the following terms with suitable explanations: **XML, JSP, MVC architecture, XSLT, XML Schema, JSTL, PHP, Database Connectivity, and Parser**. Your answer should clearly describe the purpose and role of each concept in web applications.
2. Explain how the above technologies are interconnected in building a web application. Describe how XML is structured and validated, how JSP and PHP are used for server-side processing, how MVC architecture organizes the application, and how database connectivity and parsers contribute to data processing and communication.

Code of Conduct:

*I **Abhuri Ganesh Ram Kamal 192311183** certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

A handwritten signature in blue ink, appearing to read 'Ganesh Ram Kamal', is written on a light-colored background.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Concept Understanding	25%	Clearly explains all concepts with complete accuracy and depth	Minor gaps in understanding	Basic understanding with some errors	Incorrect or very limited understanding
Content Coverage	25%	Covers all required topics comprehensively	Covers most topics with minor omissions	Covers some topics only	Very few topics covered
Technical Accuracy	20%	All explanations are technically correct and precise	Minor technical errors	Some incorrect or vague explanations	Major technical errors
Integration of Concepts	15%	Effectively connects and relates concepts logically	Some connections made with minor gaps	Limited connections between concepts	No clear relationship between concepts
Clarity & Presentation	15%	Well-structured, clear, and easy to understand	Mostly clear with minor issues	Somewhat unclear or poorly organized	Poorly structured and difficult to understand

1A. web development involves technologies that makes manage data, process user requests generate dynamic content, validate information and communicate with databases.

The following concepts are important in the development of structured and dynamic web applications.

1. XML (Extensible Markup Language)

XML is a markup language used to store and exchange structured data. Unlike HTML which is mainly designed to display information XML is designed to describe and organize data.

For example:

```
<Student>
  <name> Arun </name>
  <course> web development </course>
</Student>
```

2. JSP (Java server pages)

JSP is a Java-based server-side technology used to create dynamic web pages. A JSP page can combine HTML with Java related server-side features to generate content dynamically. When a user requests a JSP page, the web server processes it on the server and generates an HTML response that is sent to the browser.

3. MVC Architecture (Model-view-controller)

MVC is an architectural pattern that separates a web application into three major components:

Model: The model manages that application's data and business logic. It may interact with databases and represent objects such as customers, products and orders.

view: The view is responsible for presenting information to the user. Technologies such as JSP can be used to implement the view.

Controller: The controller receives user requests, processes them, communicates with the model and selects the appropriate view.

4. XSLT (Extensible Stylesheet Language Transformations)

XSLT is a language used to transform XML documents into other formats. The most common use is transforming XML into HTML for display in a web browser.

For example:

XML Data → XSLT Stylesheet → HTML output

5. XML Schema (XSD)

XML schema, commonly called XSD (XML Schema Definition), is used to define the structure and rules of an XML document.

An XML Schema can specify:

- Which elements are allowed
- Which elements are required.
- The order of element
- Data types.

6. JSTL (JavaServer Pages Standard Tag Library)

JSTL is a collection of standard tags used in JSP pages.

It allows developers to perform common operations without placing large amounts of Java code directly inside JSP pages.

JSTL supports operations such as

- Conditional processing
- variable handling
- URL handling
- Looping
- Formatting
- XML processing

7. PHP (Hypertext Preprocessor)

PHP is a server-side scripting language widely used for web development. PHP code runs on the web server and generates dynamic output usually HTML, which is sent to the user's browser.

PHP can interact with:

- Databases
- Files
- Sessions
- Forms
- APIs
- Web services

8. Database Connectivity

Database connectivity refers to the processor through which a web application communicates with a database to store, retrieve, update or delete data. In Java applications, JDBC (Java Database Connectivity) is commonly used to connect Java programs with relational databases.

9. Parser

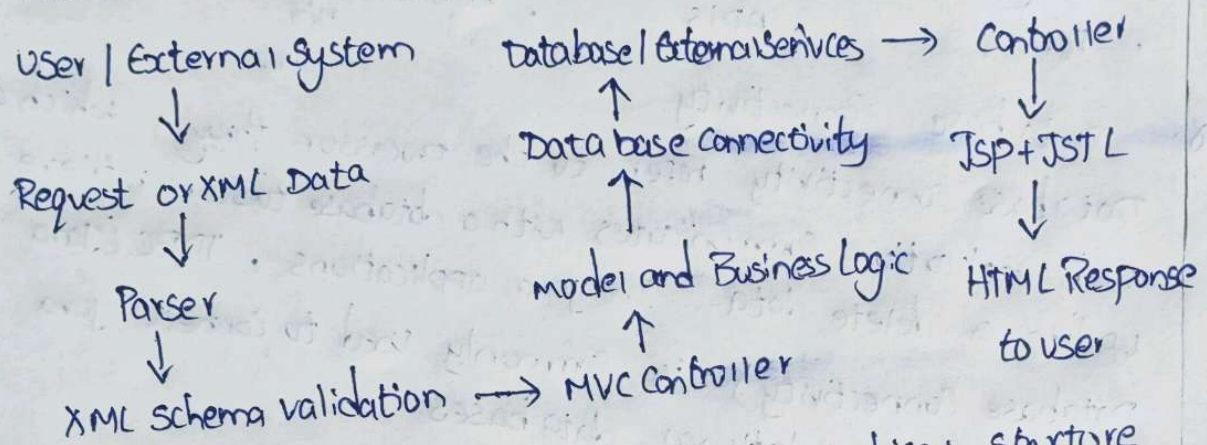
A parser is a software component that reads input data and analyzes its structure according to a defined set of rules. An XML parser, for example, reads an XML document and determines its elements, attributes, text content, and hierarchical relationships.

Common XML parsing approaches include:

- DOM Parser
- SAX Parser
- StAX Parser

2A. The technologies discussed above are interconnected because each performs a different function in the development of a web application. A typical application must receive data, process and validate it, apply business rules, communicate with databases or external systems and present results to users.

Overall Interaction in a web Application



XML is used to represent data in a hierarchical structure using custom elements and attributes. Each document normally contains a root element with child elements representing related pieces of information.

```

<order>
  <orderId> 1001 </orderId>
  <customer>
    <name> Arun </name>
    <email> arun@example.com </email>
  </customer>
  <items>
    <item>
      <productId> P101 </productId>
      <quantity> 2 </quantity>
    </item>
  </items>
</order>
  
```

<price>500</price>

</item> </items>

</order>

3. The Role of Parsers in Processing XML

Before an application can use XML data, it normally needs an XML parser to read and interpret the document.

The parser analyzes:

- Elements
- Attributes
- Text values
- Document hierarchy
- Syntax and structure.

DOM Parser

A DOM parser loads the XML document into a tree structure in memory. It is convenient when the application needs to navigate through and modify different parts of the document.

SAX Parser: A SAX parser processes XML sequentially using events. It generally uses less memory and is useful for large XML documents.

STAX Parser: A STAX parser provides a pull-based approach in which the application controls how parsing events are retrieved.

4. How XML is validated using XML Schema.

An XML document may be well-formed but still contain invalid application data. For example an order might have a missing product ID or a quantity with an incorrect data type.

XML Schema (XSD) defines rules that specify what a valid XML document should contain.

An XSD can define:

- Required elements
- optional elements
- allowed element order
- Data types
- minimum and maximum values
- allowed numbers of occurrences
- Restrictions on values and attributes.

The validation process is:

XML Document → XML parser → XML schema validation → valid

5. How MVC organizes the Application

The model-view controller (MVC) architecture organizes an application by separating its major responsibilities.

Model

The model manages application data and business rules. It may represent entities such as

- Customer
- Product
- Order
- Payment
- Inventory.

view:

The view presents information to the user. In a Java application

JSP is often used as the view. JSTL and Expression Language can help display dynamic data.